

Modifier 10.0 – Digital IC Design Hackathon

Welcome to the Challenge!

Ready to build something awesome? In **Modifier 10.0**, you'll design a **Streamed Intermediate-Result Processing Module** using **Verilog/System Verilog** that manages streamed intermediate computation results between two independent hardware blocks. It's your chance to show off your digital design skills in just 93 hours! Stick to the basics if you're new, or add cool features for extra points if you're a pro.

Main Challenge:

Design a **Streamed Intermediate-Result Processing Module** that sits between an external computation engine (which streams in partial results) and a downstream consumer (which reads out completed Output Words). Your design must:

- **Partial Transaction Reception:** Accept incoming Partial Transactions (PARTIAL_SIZE = 4 elements, DATA_WIDTH = 16 bits) on every cycle that partial_valid is asserted.
- **Processing Modes:** Support initial_partials and accumulated_partials modes within each Processing Range, including forward and reverse segment traversal.
- **Control Signal Hierarchy:** Correctly interpret segment_step, phase_change, next_range, and operation_done according to their hierarchical relationship.
- **Output Generation:** Produce completed Output Words (OUTPUT_SIZE = 8 elements) and expose them through output_addr, output_valid, and output_data.
- **Continuous Streaming:** Support concurrent input reception, intermediate data management, and output delivery.
- **Buffering:** Provide storage for at least 200 simultaneously retrievable Output Words (MAX_SEGMENTS_PER_RANGE = 100).

Optional Features (for Extra Points):

- **Output Retention Window:** Guarantee finalized Output Words remain retrievable for a downstream delay of up to 1200 clock cycles (MAX_DOWNSTREAM_DELAY).
- **Completion Latency:** Finalize all remaining Output Words within 4 clock cycles after operation_done is asserted.
- **Parameterization:** Support alternate DATA_WIDTH / PARTIAL_SIZE / OUTPUT_SIZE configurations while keeping OUTPUTS_PER_SEGMENT an integer.

Rules:

- Use **Verilog/ System Verilog** only. No VHDL.
- Simulate with **QuestaSim**. Show results with waveforms.
- **Starts: Wednesday**, July 22, 2026, 3:00 AM.
- **Ends: Saturday**, July 25, 2026, 11:59 PM (93 hours).
- Late submissions lose 5% per hour.
- No technical questions during the hackathon. Non-technical questions via email or the WhatsApp group.

Deliverables:

Submit a RAR file named **<Your_Team_Name>.rar** (e.g., TeamOne.rar) with:

1. Verilog Files

2. QuestaSim Do File

3. Simulation Results (PDF):

<Your_Team_Name>_Simulation_Results.pdf with:

- Waveform screenshots from QuestaSim showing the **simulation results**

4. Design Brief:

<Your_Team_Name>_Design_Brief.pdf (1–2 pages) explaining:

- How your design works.
- How you handled three tricky challenges.
- How you tested edge cases to show your code is original.

Submission:

Deadline: July 25, 2026 at 11:59 PM

Submission Form: <https://forms.gle/oBcm454pKQtE6CDt5>

Note: **Set the Drive link to Anyone with the link.**